

Features

- Thick film
- High voltage
- Wide resistance range
- RoHS compliant*
- UL/IEC 60950 & 60065 compatible
- UL 1676 recognized

Applications

- High voltage applications
- Consumer electronics

CHV Series - Thick Film High Voltage Chip Resistors

Electrical Characteristics

Specification		Model				
		CHV0603	CHV0805	CHV1206	CHV2010	CHV2512
Power Rating @ 70 °C		0.1 W	0.125 W	0.25 W	0.5 W	1.0 W
Operating Temperature Range		-55 °C to +155 °C				
Maximum Working Voltage		200 V	400 V	800 V	2000 V	3000 V
Maximum Overload Voltage		400 V	800 V	1600 V	3000 V	4000 V
Resistance Range	1 % E-96 + E-24	100 kΩ ~ 10 MΩ				
	5 % E-24	100 kΩ ~ 22 MΩ		100 kΩ ~ 100 MΩ		
Temperature Coefficient	1 %	±100 PPM/°C				
	5 %	±200 PPM/°C				

Environmental Characteristics

Test	Conditions	Specification
Short Time Overload	5 times rated power or max overload voltage for 5 seconds	Δ R ≤ ± (2 % + 0.1 Ω)
Solderability	+245 ±5 °C for 3 ± 0.5 seconds	Over 95 % coverage
Resistance to Solder Heat	+260 ±5 °C for 10 ±1 seconds	Δ R ≤ ± (1 % + 0.1 Ω)
Load Life Humidity	+40 ±2 °C, 90~95 % 1.5 hours ON, 0.5 hours OFF for 1000 hours at rated power	Δ R ≤ ± (5 % + 0.1 Ω)
Load Life	+70°C 1.5 hours ON, 0.5 hours OFF for 1000 hours at rated power	Δ R ≤ ± (5 % + 0.1 Ω)
Temperature Cycle	-55 °C (30 minutes), +25 °C (2~3 minutes), +155 °C (30 minutes), +25 °C (2~3 minutes) for five cycles	Δ R ≤ ± (5 % + 0.05 Ω)
Voltage Coefficient of Resistance (VCR)	Max. Test Voltage: 500 V VL: 10 % RCWV or Max. RCWV VH: 100 % RCWV or Max. RCWV	R ≤ 1 MΩ: ±100 ppm/V 1 MΩ < R < 10 MΩ: ±200 ppm/V R ≥ 10 MΩ: ±300 ppm/V

Additional Information

Click these links for more information:

PRODUCT SELECTOR | TECHNICAL LIBRARY | INVENTORY | SAMPLES | CONTACT

Agency Recognition

Description	
UL1676	File Number: E466353

How to Order

CHV 2512 - F X - 1000 E LF

Model
(CHV = Thick Film High Voltage Chip Resistor

Size
• 0603 • 0805 • 1206 • 2010 • 2512

Resistance Tolerance
F = ±1 % (Use with "X" TCR Code)
J = ±5 % (Use with "W" TCR Code)

TCR
X = ±100 PPM/°C
W = ±200 PPM/°C

Resistance Value
1 % Tolerance: First three digits are significant, fourth digit represents the number of zeroes to follow
5 % Tolerance: First two digits are significant, third digit represents the number of zeroes to follow

Packaging
E = Paper tape:
• 5,000 pcs. on 7 ″ plastic reel (CHV0603, CHV0805, CHV1206)
• 4,000 pcs. on 7 ″ plastic reel (CHV2010, CHV2512)

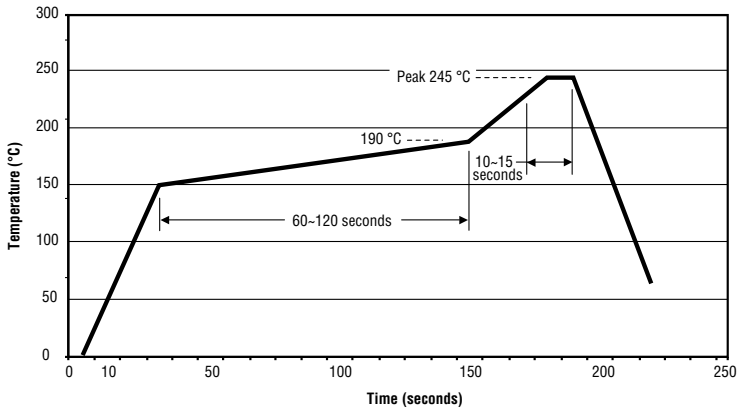
Termination
LF = Tin-plated (RoHS compliant)

CALIFORNIA WARNING: Can expose you to lead, a carcinogen and reproductive toxicant. See www.P65Warnings.ca.gov

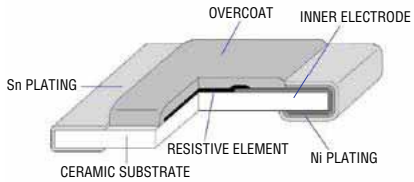
* RoHS Directive 2015/863, Mar 31, 2015 and Annex.
** Bourns® products have not been specifically designed and tested for FDA Class III applications and their use in such applications is neither recommended nor supported.

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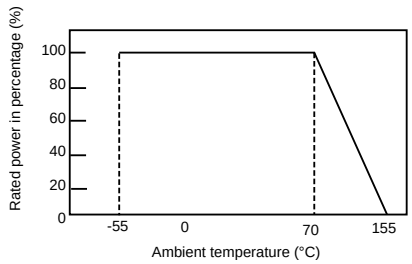
Soldering Profile



Construction

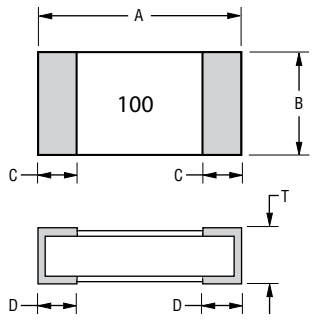


Derating Curve



Product Dimensions

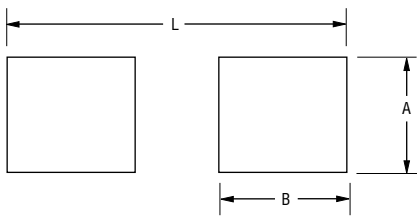
Dim.	Model				
	CHV0603	CHV0805	CHV1206	CHV2010	CHV2512
A	1.60 ± 0.10 (0.063 ± 0.004)	2.00 ± 0.10 (0.079 ± 0.004)	3.10 ± 0.10 (0.122 ± 0.004)	5.00 ± 0.20 (0.197 ± 0.008)	6.40 ± 0.20 (0.252 ± 0.008)
B	0.80 ± 0.10 (0.031 ± 0.004)	1.25 ± 0.10 (0.049 ± 0.004)	1.60 ± 0.10 (0.063 ± 0.004)	2.50 ± 0.20 (0.098 ± 0.008)	3.20 ± 0.20 (0.126 ± 0.008)
C	0.30 ± 0.20 (0.012 ± 0.008)	0.40 ± 0.20 (0.016 ± 0.008)	0.50 ± 0.20 (0.020 ± 0.008)	0.65 ± 0.25 (0.026 ± 0.010)	0.65 ± 0.25 (0.026 ± 0.010)
D	0.30 ± 0.20 (0.012 ± 0.008)	0.40 ± 0.20 (0.016 ± 0.008)	0.50 ± 0.20 (0.020 ± 0.008)	0.60 ± 0.25 (0.024 ± 0.010)	0.90 ± 0.25 (0.035 ± 0.010)
T	0.45 ± 0.10 (0.018 ± 0.004)	0.50 ± 0.10 (0.020 ± 0.004)	0.55 ± 0.10 (0.022 ± 0.004)	0.60 ± 0.10 (0.024 ± 0.004)	0.60 ± 0.15 (0.024 ± 0.006)



DIMENSIONS: MM / (INCHES)

Recommended Land Pattern

Dim.	Model				
	CHV0603	CHV0805	CHV1206	CHV2010	CHV2512
A	0.90 (0.035)	1.30 (0.051)	1.80 (0.071)	3.00 (0.118)	3.70 (0.146)
B	1.00 (0.039)	1.15 (0.045)	1.30 (0.051)	1.50 (0.059)	1.60 (0.063)
L	3.00 (0.118)	3.50 (0.138)	4.70 (0.185)	6.80 (0.268)	7.60 (0.299)



DIMENSIONS: MM / (INCHES)

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CHV Series - Thick Film High Voltage Chip Resistors

BOURNS®

Resistor Markings

CHV0603
CHV0805
CHV1206
CHV2010
CHV2512



3-Digit
E-24 ±5 % Marking

30 X 10¹
Value = 300 ohms

CHV0805
CHV1206
CHV2010
CHV2512



4-Digit
E-96/E-24 Marking

154 X 10²
Value = 15.4K ohms

CHV0603



3-Digit
E-24 ±1 % Marking

222 X 10²
Value = 2.2K ohms

CHV0603



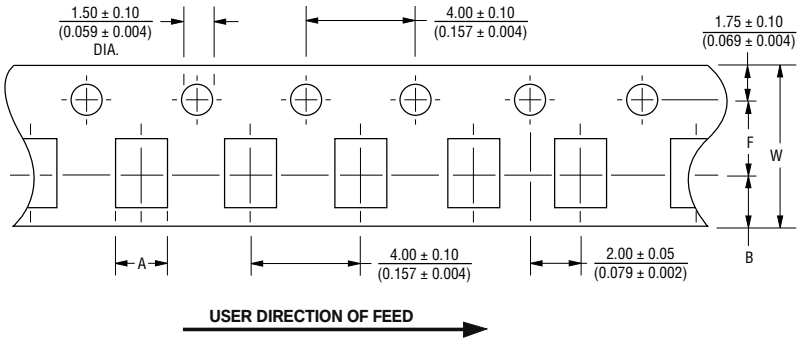
3-Digit
E-96 ±1 % Marking

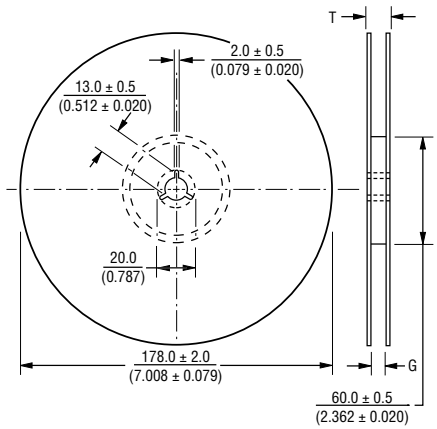
10 X 10⁰
Value = 10 ohms

Marking Explanation

- The chip color is red to identify high voltage product.
- 1 % Tolerance: 4 digits, first three digits are significant, fourth digit represents the number of zeros to follow.
- 5 % Tolerance: 3 digits, first two digits are significant, third digit represents the number of zeros to follow.

Packaging Dimensions - Tape





DIMENSIONS: MM / (INCHES)

Dim.	Model				
	CHV0603	CHV0805	CHV1206	CHV2010	CHV2512
A	1.10 ± 0.20 (0.043 ± 0.008)	1.60 ± 0.20 (0.063 ± 0.008)	2.00 ± 0.20 (0.079 ± 0.008)	2.80 ± 0.20 (0.110 ± 0.008)	3.50 ± 0.20 (0.138 ± 0.008)
B	1.90 ± 0.30 (0.075 ± 0.012)	2.40 ± 0.30 (0.094 ± 0.012)	3.57 ± 0.30 (0.141 ± 0.012)	5.50 ± 0.30 (0.217 ± 0.012)	6.70 ± 0.30 (0.264 ± 0.012)
W	8.00 ± 0.05 (0.315 ± 0.002)	8.00 ± 0.05 (0.315 ± 0.002)	8.00 ± 0.05 (0.315 ± 0.002)	12.00 ± 0.05 (0.472 ± 0.002)	12.00 ± 0.05 (0.472 ± 0.002)
F	3.50 ± 0.05 (0.138 ± 0.002)	3.50 ± 0.05 (0.138 ± 0.002)	3.50 ± 0.05 (0.138 ± 0.002)	5.50 ± 0.05 (0.217 ± 0.002)	5.50 ± 0.05 (0.217 ± 0.002)
G	10.0 ± 1.5 (0.394 ± 0.059)	10.0 ± 1.5 (0.394 ± 0.059)	10.0 ± 1.5 (0.394 ± 0.059)	13.8 ± 1.5 (0.543 ± 0.059)	13.8 ± 1.5 (0.543 ± 0.059)
T	14.9 (0.587)	14.9 (0.587)	14.9 (0.587)	16.7 (0.657)	16.7 (0.657)

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